Q1. Describe pathophysiology, clinical features, diagnosis, and management of Gastro-oesophageal Reflux Disease (GERD).

Q2. Describe aetiopathogenesis, clinical features and treatment of peptic ulcer.

Q3. Glossitis

Q4. Discuss ateiology, clinical features and management of Malabsorption Syndromes

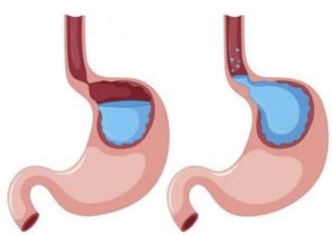
Q5. Diarrhoea

Q6. Dysphagia

Q7. Constipation

Q8. Write a short note on Acute Pancreatitis.

Q1. Describe pathophysiology, clinical features, diagnosis, and management of Gastro-oesophageal Reflux Disease (GERD).



- **Definition**: Reflux of gastric contents into the lower oesophagus.
- **Prevention Mechanisms**:
  a. Lower oesophageal sphincter (LES)
  b. Striated muscles of diaphragmatic crura
  c. Intra-abdominal pressure reinforces LES tone
  d. Oblique entry of oesophagus into stomach
  e. Swallowed saliva neutralizes refluxed acid
- Failure of these mechanisms causes **reflux oesophagitis**

Aetiology

- **Sliding hiatus hernia**: Loss of obliquity of oesophageal entry and intra-abdominal pressure reinforcement
- **Cardiomyotomy and vagotomy**: Reduce LES efficiency
- **Increased intra-abdominal pressure**: Pregnancy, obesity, ascites, straining
- **Reduced LES tone**: Smoking, alcohol, fatty foods, caffeine
- **Impaired gastric emptying**: Gastric outlet obstruction, large meals, anticholinergic drugs
- **Systemic sclerosis**
- **Drugs reducing LES tone**: Aminophylline, beta-agonists, nitrates, calcium channel blockers

Clinical Features

- **Heartburn**: Burning pain behind sternum, worse after meals, on lying down, relieved by sitting up
- **Regurgitation**: Gastric contents into mouth (acid eructation)
- **Tracheal aspiration**: Cough, laryngismus, aspiration pneumonia
- **Odynophagia**: Painful swallowing
- **Transient dysphagia**: Due to oesophageal spasm
- **Persistent dysphagia**: Due to strictures
- **Iron deficiency anaemia**: From blood loss

Complications

- **Oesophagitis**
- **Oesophageal strictures**
- **Oesophageal ulcers**
- **Aspiration pneumonia**
- **Barrett's oesophagus**
- **Carcinoma of oesophagus**

Investigations

- **Endoscopy**: Identifies oesophagitis, strictures, Barrett's mucosa; biopsy confirms findings
- **Barium meal**: Detects hiatus hernia
- **Bernstein test**: For high suspicion but negative endoscopy
- **ECG**: Rules out ischaemic heart disease
- **Oesophageal motility studies**

Treatment

General Measures

- Weight reduction
- Small frequent meals
- Avoid smoking, alcohol, fatty food, caffeine, mint, medications
- Avoid late night meals
- Avoid lifting weights, stooping, bending
- Elevate head end of the bed to 15°

Medical Treatment

- **Mild cases**: Liquid antacid 10-15 mL, 1 and 3 hrs after meals
- **Moderate cases**: H_2 receptor antagonists (Cimetidine 400 mg/Ranitidine 150 mg b.d/q.i.d) for 6 weeks
- **Severe cases**: Proton Pump Inhibitors (Omeprazole 20-40 mg, Pantoprazole 40 mg, etc.) for 6-8 weeks
- **Adjunctive therapy**: Metoclopramide/Domperidone 10 mg t.i.d to increase LES tone and gastric emptying
- **Stricture**: Repeated dilatations, iron or blood transfusion for anaemia

Surgical Treatment

- **Stricture resection**
- **Sliding hiatus hernia**: Fundoplication (additional valve mechanism) to return LES to abdomen

Q2. Describe aetiopathogenesis, clinical features and treatment of peptic ulcer.

- **Definition**: Ulcer in lower oesophagus, stomach, duodenum, jejunum (post-surgical anastomosis), or ileum near Meckel's diverticulum
- **Cause**: Imbalance between mucosal defense and luminal factors (acid and pepsin)
- **Characteristics**: Breach >5 mm deep into submucosa

- **Common Types**:
  a. Duodenal ulcer: Common, affects younger age (30-55 years)
  b. Gastric ulcer: Less common, affects older age (55-70 years)

Aetiology

- **Hereditary**: Strong family history, especially in gastric ulcers
- **Helicobacter pylori**:
  a. Main cause (70% gastric ulcers)
  b. Produces mucosal damage via urease, catalase, lipase, adhesins
  c. Converts urea to ammonia, causing mucosal damage
- **NSAIDs**:
  a. Decrease prostaglandin levels, impair cytoprotection
  b. Cause mucosal injury, erosions, ulcers
- **Smoking**: Delays healing, increases complications
- **Corticosteroids**: Silent perforation
- **Acid-Pepsin vs. Mucosal Resistance**:
  a. Gastric hypersecretion (common in Zollinger-Ellison syndrome)
  b. Acid important in duodenal ulcers
- **Factors Reducing Mucosal Resistance**:
  a. Drugs (aspirin, NSAIDs)
  b. H. pylori
  c. Bile reflux from weak pyloric sphincter
- **Other Risks**: Smoking, alcohol

Clinical Features

- **Chronic Condition**: Relapses and remissions over years
- **Abdominal Pain**:
  a. Localized to epigastrium (pointing sign)
  b. Burning, hunger pain (relieved by food/antacids)
  c. Night pain (wakes at 3 AM, relieved by food)
  d. Episodic pain (lasts 1-3 weeks, occurs 3-4 times a year)
- **Seasonal Relapses**: More in winter and spring
- **Other Symptoms**:
  a. Water-brash, heartburn, vomiting, nausea, dyspepsia
  b. Haematemesis (more common), weight loss, anaemia

Complications

- Haematemesis
- Gastric outlet obstruction
- Acute perforation

- Chronic blood loss anaemia

Investigations

- **Barium Meal**: Shows ulcer as a crater or deformed cap
- **Endoscopy**: Visualizes ulcer, biopsy rules out malignancy and H. pylori
- **Serum Gastrin and Gastric Acid**: For Zollinger-Ellison syndrome
- **H. pylori Tests**:
  a. Invasive: Rapid urease test, histology, culture from biopsy
  b. Non-invasive: Serology, urea breath test

Treatment of Peptic Ulcer

I. Short-term Management

Drugs used:

1. **Antacids & Alginates**
   - Examples: Aluminium hydroxide, magnesium trisilicate, sodium bicarbonate.
   - **Action**: Forms a protective mucosal "raft".
   - **Dose**: 15-30 ml liquid after meals & at bedtime for 4-6 weeks.
   - **Use**: Symptomatic relief.
2. **H2-receptor Antagonists**
   - Examples: Cimetidine, ranitidine, famotidine.
   - **Action**: Inhibit H_2 receptors on parietal cells.
   - **Dose**:
     - Cimetidine: 400 mg BD or 800 mg at night.
     - Ranitidine: 150 mg BD or 300 mg at night.
     - Famotidine: 20 mg BD or 40 mg at night.
   - **Duration**:
     - Duodenal ulcer: 4 weeks.
     - Gastric ulcer: 6 weeks + follow-up endoscopy.
3. **Proton Pump Inhibitors (PPIs)**
   - Examples: Omeprazole, lansoprazole, pantoprazole, esomeprazole, rabeprazole.
   - **Action**: Irreversibly inhibits H+/K+ ATPase (proton pump).
   - **Dosage**:
     - Omeprazole, rabeprazole: 20 mg daily.
     - Lansoprazole: 15-30 mg daily.
     - Esomeprazole: 20-40 mg daily.
     - Pantoprazole: 40 mg daily.
   - **Indications**: GERD, Zollinger-Ellison syndrome, H. pylori adjunct therapy.
   - **Note**: Take 30 mins before meals (except rabeprazole/pantoprazole).
4. **Prostaglandin Analogues**

- Example: Misoprostol (200 mcg QID).
- **Action**: Enhances mucosal protection.
- **Use**: Prevent NSAID-induced ulcers.

5. **Colloidal Bismuth Compounds**
   - Examples: Bismuth subsalicylate, bismuth subcitrate.
   - **Action**:
     - Forms protective coat over ulcer.
     - Antimicrobial against H. pylori.
6. **Complex Salts**
   - Example: Sucralfate.
   - **Action**: Coats ulcers, protects from acid-pepsin digestion.

H. Pylori Eradication Therapy

- **Triple Therapy**:
  - PPI + Amoxicillin (1 g BD) + Clarithromycin (500 mg BD) for 14 days.
  - Metronidazole replaces amoxicillin if allergic.
- **Quadruple Therapy** (if triple therapy fails):
  - PPI + Bismuth + Tetracycline + Metronidazole.

II. Long-term Management

1. **Intermittent Treatment**:
   - For relapses <4 times/year.
   - Use ulcer healing agents for 4 weeks.
2. **Maintenance Treatment**:
   - Rarely needed post-H. pylori eradication.
   - Lowest effective dose of H_2 blockers if required:
     - Cimetidine: 400 mg at night.
     - Ranitidine: 150 mg at night.
3. **Surgical Treatment**:
   - **Gastric ulcer**: Partial gastrectomy (Billroth I anastomosis).
   - **Duodenal ulcer**: Truncal vagotomy + pyloroplasty or gastroenterostomy.
   - **Emergency surgery**: For bleeding or perforation.
   - **Elective surgery**: For gastric outlet obstruction or recurrent ulcers.

Q3. Glossitis

Definition

- Glossitis is an abnormality of the tongue caused by acute or chronic inflammation.
- It causes **swelling** and a **change in colour** of the tongue.
- **Papillae** are lost, making the tongue appear **smooth**.

Etiology

Local Causes:

- **Bacterial or viral infections**.
- **Mechanical irritation** or injury from burns, rough edges of teeth, dental appliances, or trauma.
- **Exposure to irritants**: Tobacco, alcohol, hot foods, spices.
- **Sensitization (allergic reaction):** Toothpaste, mouthwash, breath fresheners, dyes in candy, or plastic in dentures/retainers.

Systemic Causes:

- **Iron deficiency** and **pernicious anaemia**.
- **Vitamin B deficiencies**.
- **Oral lichen planus**.
- **Erythema multiforme**.
- **Aphthous ulcers**.
- **Pemphigus vulgaris**.
- **Syphilis** and other disorders.
- In some cases, glossitis can be **inherited**.

Clinical Features

- Tongue appears **swollen** with a **smooth** surface.
- Tongue colour:
  - **Dark beefy red**.
  - **Pale** in pernicious anaemia.
  - **Fiery red** in vitamin B deficiency.
- Tongue may be **sore** and **tender**.

- **Absence of papillae** (nodules on the tongue's surface).

Investigations

- Laboratory tests to confirm **systemic causes**.

Treatment

- Goal: **Reduce inflammation**.
- **Oral hygiene**: Brush teeth **twice a day** and floss daily.
- **Corticosteroids** (e.g., prednisone) may be used to reduce inflammation.
- For mild cases, **topical prednisone suspension** can be used as a mouth rinse.
- **Antibiotics, antifungals**, or other **antimicrobials** if infection is the cause.
- **Treat anaemia** and **nutritional deficiencies** with dietary changes or supplements.
- **Avoid irritants**: Hot, spicy foods, alcohol, tobacco to minimize discomfort.

Complications

- **Discomfort**.
- **Airway blockage**.
- Difficulty with **speaking**, **chewing**, or **swallowing**.

Q4. Discuss ateiology, clinical features and management of Malabsorption Syndromes

Definition

- Disorders of digestion and diminished absorption of dietary nutrients.
- Various diseases can lead to malabsorption.

Aetiology
Disorders of Intraluminal Digestion:
A. **Pancreatic enzyme deficiency**:

- **Chronic pancreatitis**.

- **Cystic fibrosis**.
- **Pancreatic carcinoma**.

B. **Disturbances of gastric function**:

- **Gastroenterostomy**.
- **Partial gastrectomy**.

C. **Deficiency of bile acids**:

- **Crohn's disease**.
- **Resection of terminal ileum**.
- **Stagnant loop syndrome**.

Disorders of Transport in the Intestinal Mucosal Cell:
A. **Histologically abnormal mucosa**:

- **Lymphoma**.
- **Coeliac disease**.
- **Tropical sprue**.
- **Whipple's disease**.
- **Giardiasis**.
- **Radiation enteritis**.

B. **Histologically normal mucosa** (genetic diseases):

- **Lactase deficiency**.
- **Pernicious anaemia**.

C. **Disorders of transport from mucosal cell**:

- **Abdominal lymphoma**.
- **Tuberculosis**.
- **Telangiectasia of mesenteric lymphatics**.
- **Abetalipoproteinaemia**.

D. **Impaired nutrient uptake**:

- **Lymphatic obstruction**.
- **CHF** (Congestive heart failure), **pericarditis**.

E. **Miscellaneous**:

- **Diabetes mellitus**.
- **Hyperthyroidism**.

- **Hypoparathyroidism**.

Clinical Features
General Features:

- **Diarrhoea**.
- **Abdominal pain**.
- **Distension**.
- **Weight loss**.
- **Anaemia**.
- **Vague ill health**.

Specific Features:
i. **Carbohydrates**:

- **Abdominal distension**.
- **Belching**.
- **Bloating**.

ii. **Protein**:

- **Progressive emaciation**.
- **Pitting pedal oedema**.

iii. **Fat**:

- **Weight loss**.
- **Diarrhoea**.
- **Loose, pale, bulky, offensive stools** (steatorrhoea).

iv. **Vitamins**:

- **Vitamin A**: **Follicular keratosis**, **night blindness**, **xerophthalmia**, **keratomalacia**.
- **Vitamin D**: **Muscular irritability**, **tetany**, **osteomalacia**.
- **Vitamin K**: **Haemorrhagic tendencies**.
- **Vitamin B_1 and B_2**: **Angular stomatitis**, **cheilosis**, **glossitis**, **neuropathy**.
- **Folic acid**: **Macrocytic anaemia**, **glossitis**.

v. **Minerals and Electrolytes**:

- **Sodium**: **Muscle cramps**, **weakness**, **hypotension**.
- **Potassium**: **Weakness**, **areflexia**, **intestinal distension**, **cardiac arrhythmias**.
- **Calcium**: **Muscular irritability**, **tetany**, **rickets**, **osteomalacia**.
- **Magnesium**: **Weakness**, **tingling sensation**, **tetany**.
- **Zinc**: **Anorexia**, **weakness**, **tingling**, **impaired taste**.

- **Iron**: **Hypochromic microcytic anaemia**, **glossitis**, **koilonychia**.
- **Water**: **Dehydration**, **low blood volume**.

Investigations
Routine Laboratory Studies:

- Tests to detect nutrient deficiency.
- Do not establish the cause but reveal specific malabsorption.

Specific Tests:
i. **Faecal fat estimation**:

- Confirms **steatorrhoea** and fat malabsorption.
- **Sudan III stain** may show increased stool fat.
- **72-hour stool collection**: >10g fat/day suggests fat malabsorption.

ii. **Schilling test**:

- For **cobalamin (B12) malabsorption**.
- **Radio-labelled cobalamin** is given, and excretion is measured.
- **<10% excretion** in 24 hours is abnormal.
- Repeated with intrinsic factor or pancreatic enzymes for further diagnosis.

iii. **D-xylose test**:

- Detects **carbohydrate malabsorption**.
- **Excretion <4.5g** in 5 hours indicates malabsorption.

iv. **Upper GI endoscopy and biopsy**:

- Essential for diagnosing conditions like **tropical sprue**, **celiac sprue**, **Whipple's disease**, and **Crohn's disease**.

v. **Barium meal contrast radiography**:

- Evaluates structural abnormalities like those in **Crohn's disease**, **diverticulae**, and **strictures**.

vi. **Pancreatic exocrine functions**:

- Assessed in **steatorrhoea** cases.

vii. **Serological studies**:

- Detect **autoantibodies** in conditions like **celiac sprue** and **pernicious anaemia**.

viii. **Breath tests**:

- **Cholyl-14C-glycine** and **Lactose H2** for bacterial overgrowth.

ix. **Small intestinal biopsy**:

- **Duodenal or jejunal biopsy** to confirm diagnosis.

Treatment

- **Replace deficient nutrients**.
- **Gluten-free diet** in **coeliac disease**.
- **Pancreatic supplements** for pancreatic insufficiency.
- **Low-fat diet** and **cholestyramine** for **bile acid deficiency**.
- **Replacement therapy** for **anaemia**, **bone disease**, and **coagulation defects**.
- **Oral folic acid**, **oral iron**, and **intramuscular B12**.
- **Vitamin D** and **calcium supplements.**
- **Vitamin B complex**.
- **Treat dehydration** and **electrolyte deficiency** with **intravenous infusion**.

Q5. Diarrhoea

Acute Diarrhoea

Acute diarrhoea is mainly caused by infections (90%). It can also result from drugs, ischaemia, toxins, and other conditions.

Aetiology: Causes of acute diarrhoea include:

A. Infectious: i. **Viral:**

- **Rotavirus**
- **Norwalk agents**
- **Cytomegalovirus**

ii. **Bacterial:**

- **Preformed toxin:**
  - **S. aureus**
  - **B. cereus**
  - **Clostridium perfringens**
- **Enterotoxin induced:**
  - **Enterotoxigenic E. coli (ETEC)**
  - **Vibrio cholerae**
- **Cytotoxin production:**

    - **Enterohaemorrhagic E. coli (EHEC)**
    - **Clostridium difficile**
- **Mucosal invasion:**
    - **Shigella**
    - **Campylobacter**
    - **Salmonella jejuni**
    - **Enteroinvasive E. coli (EIEC)**
    - **Yersinia enterocolitica**

iii. **Protozoal:**

- **Entamoeba histolytica**
- **Giardia lamblia**
- **Cryptosporidium**

B. Non-Infectious:

- **Inflammatory bowel disease (ulcerative colitis, Crohn's disease)**
- **Metabolic (DKA, carcinoid)**
- **Sepsis**
- **Gluten-free diet in coeliac disease**
- **Low fat diet and cholestyramine for bile acid deficiency**
- **Diverticulitis**
- **Replacement therapy for anaemia, bone disease, and coagulation defects**
- **Oral folic acid, oral iron, and intramuscular B12**
- **Vitamin D and calcium supplements**
- **Vitamin B complex**

Clinical Features:

- **Steatorrhoea** is the presenting symptom.
- **Diarrhoea** or **abdominal discomfort**.
- **Nutritional deficiencies**, e.g., deficiency of **vitamin A**, **D**, **B12**, and **K**.
- General features like **anaemia**, **sore mouth**, **weight loss**, **fatigue**, and **lethargy**.
- **Bone pain** may be present.
- **Skin changes** like **pellagra**.
- **Peripheral neuropathy**, **irritability**, and **lack of confidence**.

Investigations: i. **Routine laboratory studies:** Tests to detect any nutrient deficiency.
ii. **Specific tests:**

- **Faecal fat estimation**
- **Schilling test**
- **D-xylose test**

- **Upper GI endoscopy and biopsy of small intestinal mucosa**
- **Barium meal contrast radiography**
- **Pancreatic exocrine functions**
- **Serological studies**
- **Breath tests**
- **Small intestinal biopsy** (duodenal or jejunal)

Treatment:

- Replace deficient nutrients.
- **Pancreatic supplements** in **pancreatic insufficiency**.

General Management:

- **Rest** and maintain **fluid and electrolyte balance**.
- **ORS** should be given to all children as early as possible.
- **IV fluid** is required for patients with constant vomiting or moderate to severe dehydration.
- **Ringer lactate** is ideal; **normal saline** can be given.

Anti-microbial / Anti-diarrhoeal Treatment:

- **Ciprofloxacin 500 mg BD** for three days or **Nalidixic acid 1 gm 6 hourly** for 5-7 days.
- It may be combined with **Tinidazole 300 mg BD**.

Anti-motility Agents:

- Should not be used in **children below 5 years**.
- **Loperamide** or **diphenoxylate atropine**. **Codeine** could also be used.
- **Sodium and water conserving agents** like **racecadrol** can be used in children and adults. It reduces the loss of sodium and water in the stool.

Diagnosis:

- Stool culture for **bacteria** or **virus**.
- Stool study for **parasite**.

Non-Infectious Causes:

- **Drugs** like **NSAIDs**, **antibiotics**.
- **Ischaemic colitis**.
- **Inflammatory bowel disease**, **diverticulitis**, and other metabolic disorders.

Q6. Dysphagia
Difficulty or a sense of obstruction while swallowing food is known as **dysphagia**.

Causes:

a. Mechanical:

- **Luminal:** Large **bolus** (big chunk of food), **foreign body**
- **Intrinsic narrowing:** Inflammation, **webs and rings**, benign **strictures**, **malignancy** (cancer)
- **Extrinsic compression:** Conditions like **cervical spondylosis**, **retropharyngeal abscess**, **enlarged thyroid**, **aortic aneurysm**

b. Neuromuscular:

- **Difficulty in initiation:** Conditions like **Sjogren's syndrome**, paralysis of **tongue**, **oral anaesthesia**
- **Disorders of striated muscle:**
    - **Bulbar or pseudobulbar paralysis**
    - **Myasthenia gravis**
    - **Motor neuron disease**
    - **Polymyositis**
    - **Myopathies**
    - **Rabies**
    - **Tetanus**

c. Disorders of smooth muscle:

- **Scleroderma**
- **Myotonic dystrophy**
- **Oesophageal spasm**
- **Achalasia cardia**
- **Chagas disease**

Clinical Features:

- **Dysphagia to liquids:** Often due to **neuromuscular** issues.
- **Duration and course** of dysphagia help in diagnosis.
- **Associated symptoms** like **nasal regurgitation**.
- **Bulbar palsy** (weakness in muscles responsible for speech and swallowing).
- **Severe weight loss** due to difficulty eating.
- **Hoarseness of voice**
- **Malignancy (cancer)** may cause dysphagia.
- **Laryngitis** (inflammation of the voice box).
- **Chest pain** related to swallowing problems.

Investigations:

- **Barium swallow:** A test where the patient swallows a barium solution to check the oesophagus.
- **Upper GI endoscopy:** A procedure to look inside the digestive tract.
- **Oesophageal motility studies:** Tests that check how well the muscles in the oesophagus are working.

Treatment:

- **Treatment of cause:** Address the underlying cause of dysphagia.
- **Dietary modification:** Changing the type of food to make swallowing easier.
- **Nasogastric tube feeding:** A tube is inserted into the nose and down to the stomach for feeding.
- **Gastrostomy/jejunostomy:** Surgical procedures to provide long-term feeding through a tube directly into the stomach or small intestine.

Q7. Constipation

Constipation refers to having **bowel movements** less frequently than **3 times a week** or when stools are **hard** and difficult to pass.

Causes of Constipation:

Acute Causes:

- **Dehydration:** Lack of enough fluids in the body makes stools dry and hard, making them difficult to pass.
- **Acute intestinal obstruction:** A blockage in the intestines that prevents the normal flow of food and waste.
- **Acute appendicitis:** Inflammation of the appendix can cause pain and swelling, leading to difficulty in passing stools.

Chronic Causes:

I. Functional Causes:

- **Rectal stasis:** The inability of the rectum to push waste out, causing a delay in bowel movements.
- **Faulty habits:** Poor bowel habits, like ignoring the urge to go to the toilet, can lead to constipation.
- **Impaired consciousness:** Conditions that affect mental status (e.g., coma or severe illness) can prevent a person from recognizing the need to pass stools.
- **Painful anal area:** Conditions such as **hemorrhoids** or **anal fissures** make it painful to pass stools, causing a person to hold it in.

- **Colonic stasis:** The colon slows down, and stool moves slowly, causing constipation.
- **Decreased food intake:** Not eating enough food, especially foods high in fiber, can lead to less bowel movement.
- **Decreased fibre residue:** Low intake of fiber makes stool harder and more difficult to pass.
- **Endocrine dysfunction:** Problems with hormones, such as in **hypothyroidism** (underactive thyroid), can slow down bowel movements.
- **Drugs:** Certain medications like **cocaine**, **morphine**, **antidepressants**, and **calcium channel blockers** can cause constipation as a side effect.
- **Irritable bowel syndrome (IBS):** A functional disorder that affects the intestines and can lead to constipation, diarrhea, or a combination of both.

II. Organic Causes:

- **Myxoedema:** A condition caused by severe **hypothyroidism**, where the thyroid gland doesn't produce enough hormones, slowing down bodily functions, including bowel movements.
- **Systemic sclerosis:** A connective tissue disease that can lead to thickening and scarring of tissues, including the intestines, making bowel movements difficult.
- **Diabetes mellitus:** High blood sugar levels over time can affect the nerves, including those that control the bowel, leading to constipation.
- **Depression:** Psychological conditions like depression can alter bowel habits, causing slow movement of stool.
- **Hypercalcaemia:** High levels of **calcium** in the blood can affect the muscles of the intestines, leading to constipation.
- **Diverticulitis:** Inflammation of the small pouches in the wall of the colon (diverticula), which can lead to constipation and abdominal pain.
- **Megacolon:** A condition where the colon becomes abnormally dilated and loses its ability to move stool properly.
- **Pressure on the rectum:** Tumors or a **pregnant uterus** putting pressure on the rectum can prevent normal bowel movements.
- **Neurological conditions:** Diseases like **Parkinson's disease** or **spinal cord injuries** can affect the nerves that control bowel movements, leading to constipation.

Approach to a Patient with Dysphagia:

- **Dysphagia to solids** (difficulty swallowing solid food) often points to **mechanical dysphagia**, which can be caused by **obstructions** or structural problems in the esophagus.

Q8. Write a short note on Acute Pancreatitis.

- **Definition**: Acute pancreatitis is an **inflammation** of the pancreas that develops suddenly. It can range from mild discomfort to a life-threatening condition.
- **Causes**:
  - **Gallstones**: The most common cause, blocking the bile duct and obstructing the pancreas.
  - **Alcohol abuse**: Heavy drinking can lead to inflammation of the pancreas.
  - **High triglycerides**: Elevated blood fat levels can cause pancreatitis.
  - **Infections**: Certain viral or bacterial infections can lead to pancreatitis.
  - **Trauma**: Injury to the abdomen, including surgery or accidents.
  - **Medications**: Some drugs like steroids, diuretics, and certain antibiotics can cause pancreatitis.
- **Pathophysiology**:
  - The inflammation occurs when the pancreas' **digestive enzymes** (amylase, lipase) are activated prematurely inside the gland.
  - This leads to **self-digestion** of pancreatic tissue, causing damage and swelling.
- **Clinical Features**:
  - **Abdominal pain**: Often **severe**, located in the upper abdomen, and may radiate to the back.
  - **Nausea and vomiting**: Common and often persistent.
  - **Fever**: Low-grade fever due to inflammation.
  - **Jaundice**: Yellowing of the skin and eyes if the bile duct is blocked.
  - **Tachycardia**: Increased heart rate due to pain and inflammation.
- **Diagnosis**:
  - **History and clinical examination**: Identifying risk factors like alcohol consumption or recent trauma.
  - **Blood tests**: Elevated levels of **serum amylase** and **lipase** are key markers.
  - **Imaging**: **Ultrasound** and **CT scan** can help confirm diagnosis and identify complications like pancreatic necrosis.
- **Treatment**:
  - **Supportive care**:
    - **Hydration** with IV fluids to maintain blood pressure.
    - **Pain management**: Use of **analgesics** like morphine to relieve pain.
    - **Fasting**: No food or drink by mouth for the first few days to allow the pancreas to rest.
  - **Antibiotics**: Only if there's evidence of infection.
  - **Surgery**: In cases of severe pancreatitis, surgery may be needed to remove damaged tissue or gallstones.
- **Complications**:
  - **Pancreatic pseudocyst**: A fluid-filled sac that can form in the pancreas.
  - **Infection**: Pancreatic tissue can get infected, leading to sepsis.

    - **Organ failure**: In severe cases, the heart, kidneys, or lungs can fail due to systemic effects of inflammation.
    - **Chronic pancreatitis**: Repeated episodes of acute pancreatitis can lead to long-term damage and scarring of the pancreas.
- **Prognosis**:
    - The majority of patients recover fully with proper treatment, but severe cases can lead to complications and a higher risk of death, especially in the elderly or those with pre-existing conditions like diabetes.